Zero Knowledge Bootcamp

#3 & 4. Cyclic groups and elliptic curves

RareSkills

Review of groups

A group G is a set together with an operation $\oplus$ that obeys the following properties:

closure – if $a \in G$ and $b \in G \rightarrow a \oplus b \in G$
associativity – $(a \oplus b) \oplus c = a \oplus (b \oplus c)$
neutral element – $\exists\, e \in G$ s.t. $\forall a \in G, a \oplus e = e \oplus a = a$
inverse element – $\forall a \in G\, \exists\, a^{-1} \in G$ s.t. $a \oplus a^{-1} = a^{-1} \oplus a = e$

RareSkills

Cyclic groups

A group is cyclic if all its elements can be generated through a single (not necessarily unique) element.

$$\text{let } G \text{ be a group, then all } a \in G \text{ can be written as}$$
$$a = g^n, \text{where } g^n = g \oplus g \oplus g \oplus \ldots \oplus g,\ n \text{ times}$$

Where g is a called a generator of the group.

RareSkills

Example

Consider $\mathbb{Z}_5^* = \{1, 2, 3, 4\}$

$2^1 = 2$	$3^1 = 3$	$4^1 = 4$
$2^2 = 4$	$3^2 = 4$	$4^2 = 1$
$2^3 = 3$	$3^3 = 2$	$4^3 = 4$
$2^4 = 1$	$3^4 = 1$	$4^4 = 1$

RareSkills

Exponential map

In a cyclic group we can define an **exponential map** as

$$g^{(\cdot)} : \mathbb{Z}_n \to G \text{ s.t. } x \to g^x$$

$$\mathbb{Z}_4 \to \mathbb{Z}_5^*$$

$$0 \to 3^0 = 1$$

$$1 \to 3^1 = 3$$

$$2 \to 3^2 = 4$$

$$3 \to 3^3 = 2$$

RareSkills

Logarithm map

We can define the inverse of the exponential map as

$$\log : G \to \mathbb{Z}_n \text{ s.t. } g^x \to x$$

$$\mathbb{Z}_5^* \to \mathbb{Z}_4$$

$$3^0 = 1 \to 0$$

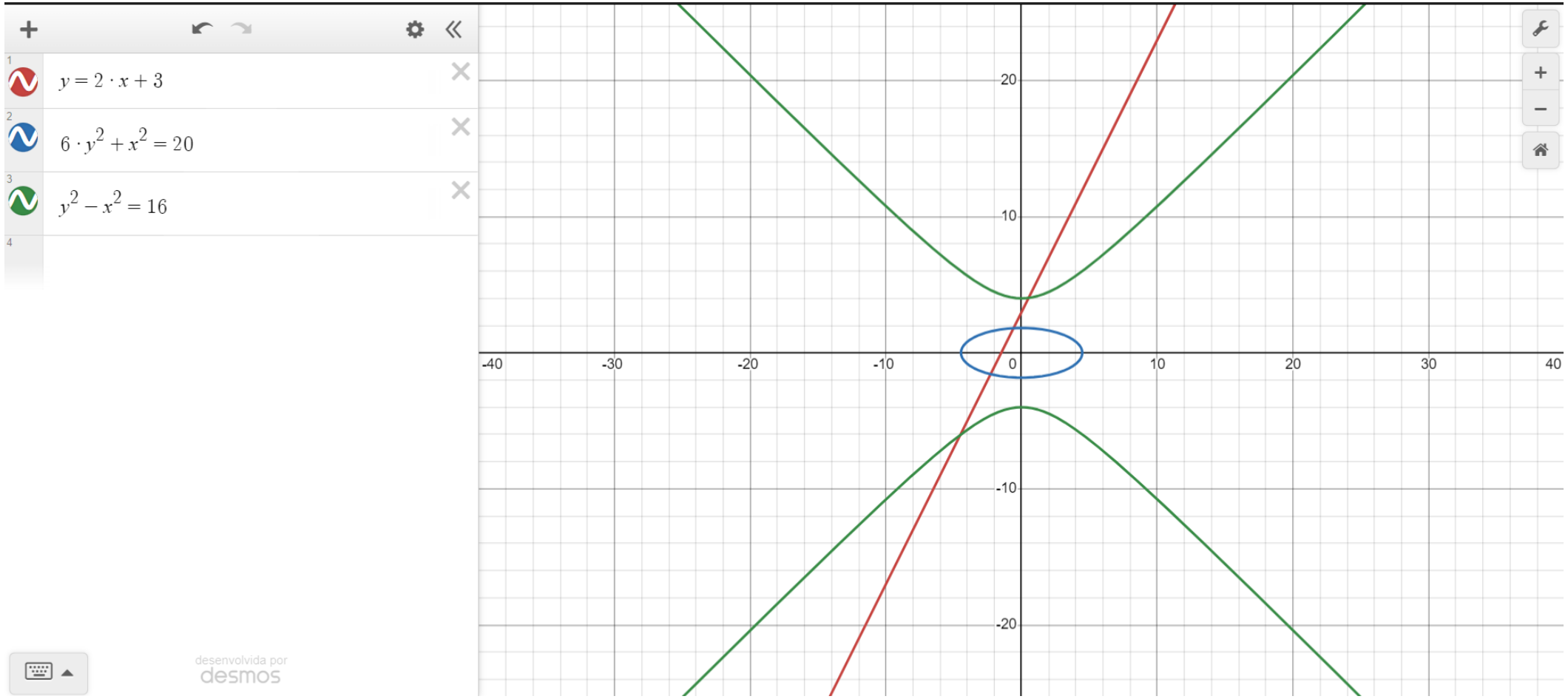
$$3^1 = 3 \to 1$$

$$3^2 = 4 \to 2$$

$$3^3 = 2 \to 3$$

RareSkills

Equations and curves



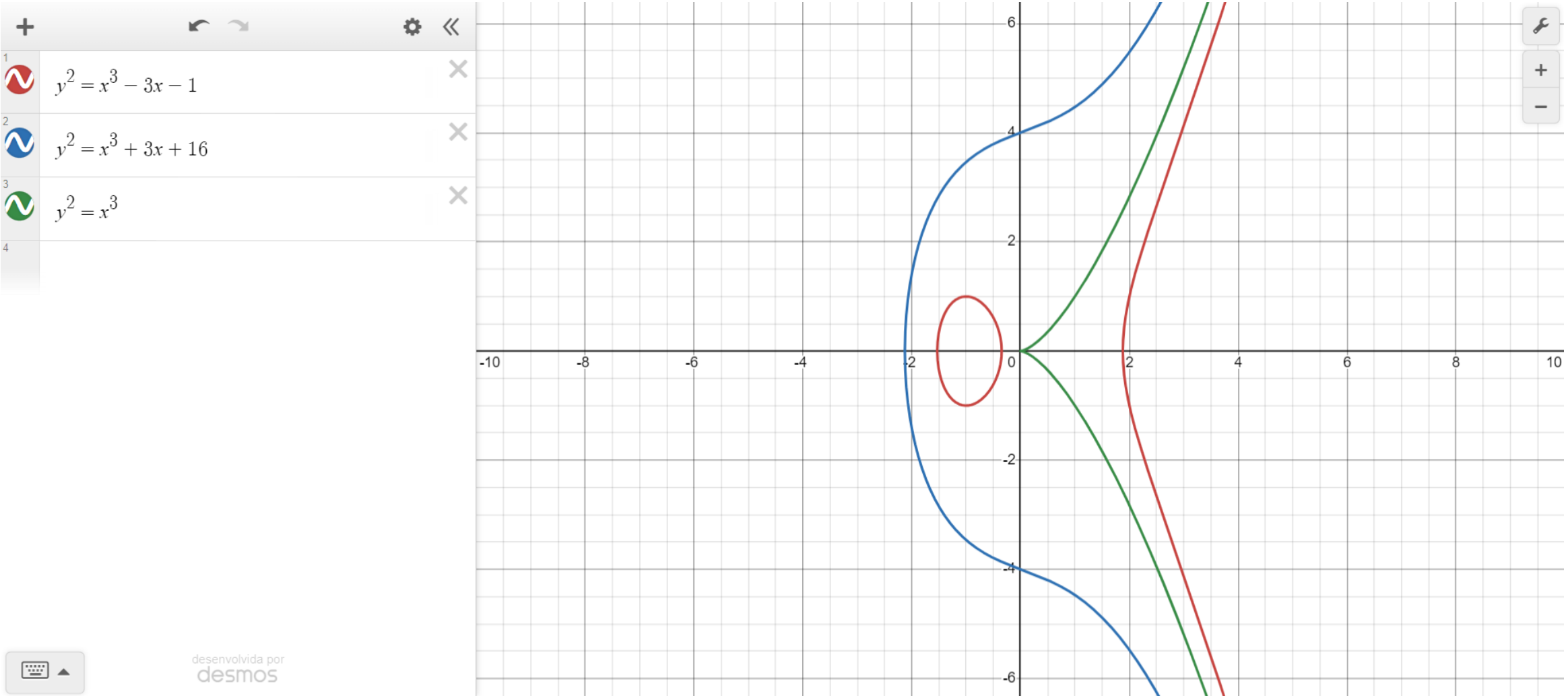
RareSkills

Short Weierstrass equation

$$y^2 = x^3 + ax + b$$

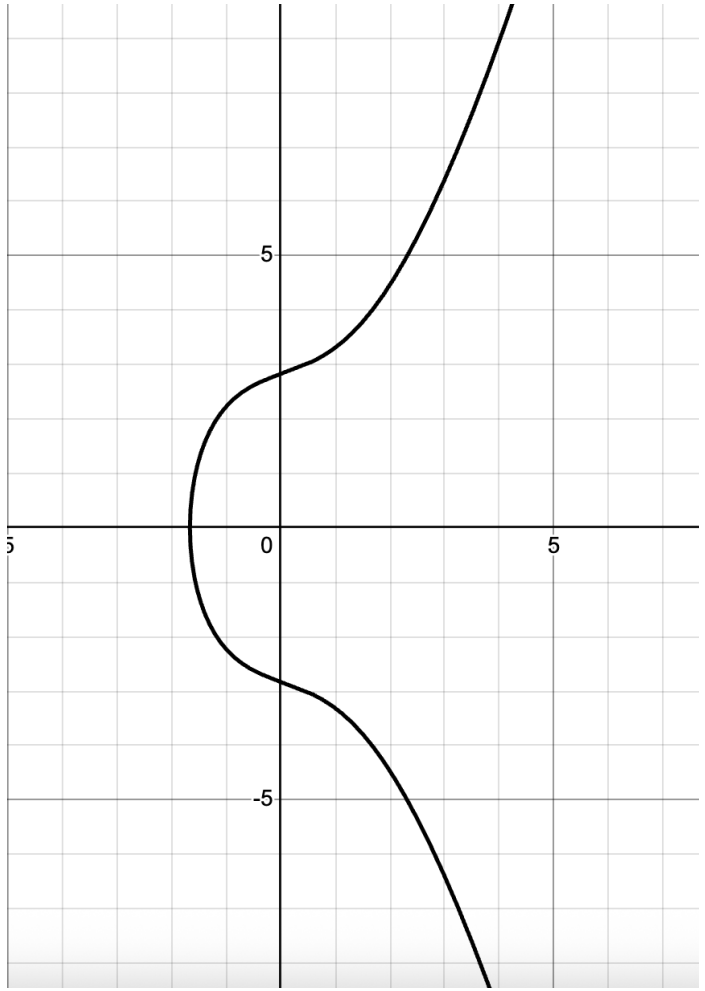
$$4a^3 + 27b^2 \neq 0$$

RareSkills



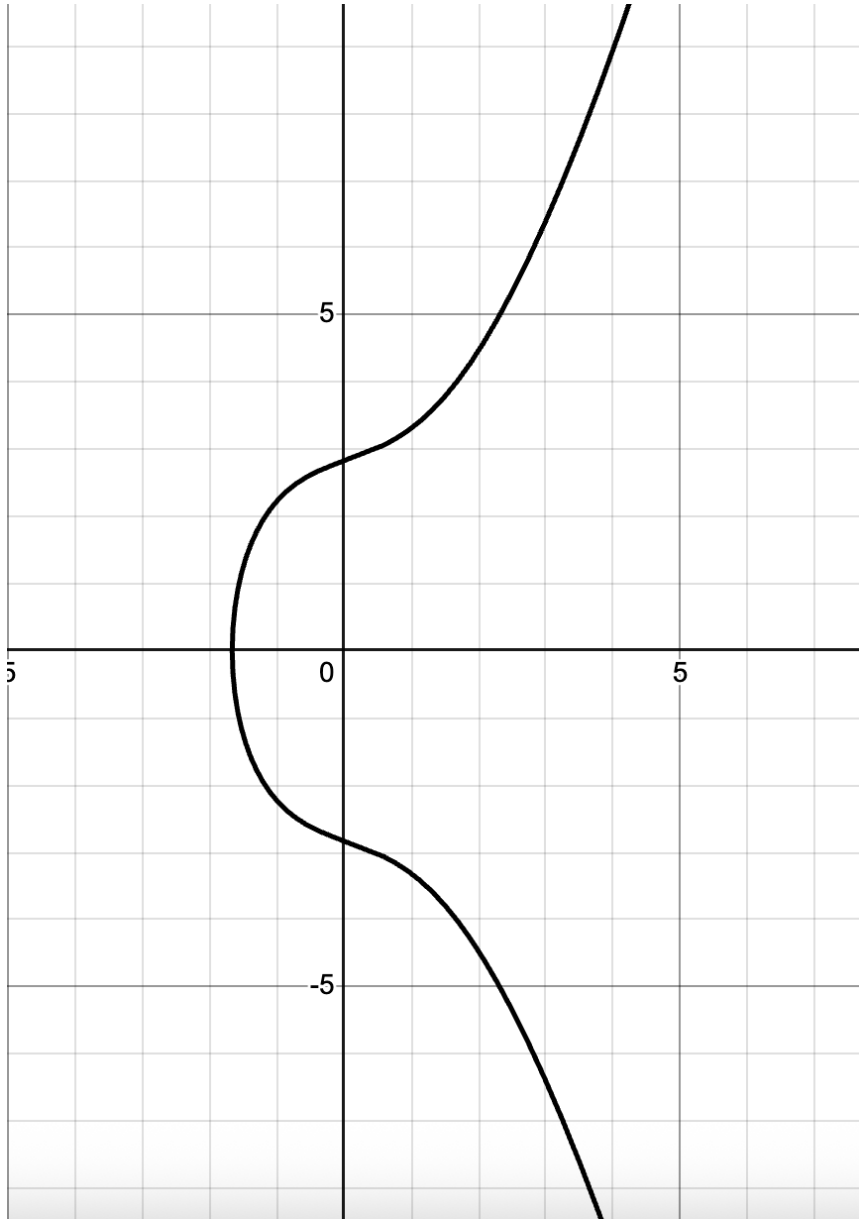
RareSkills

From 2 solutions (points) we can find a third one...



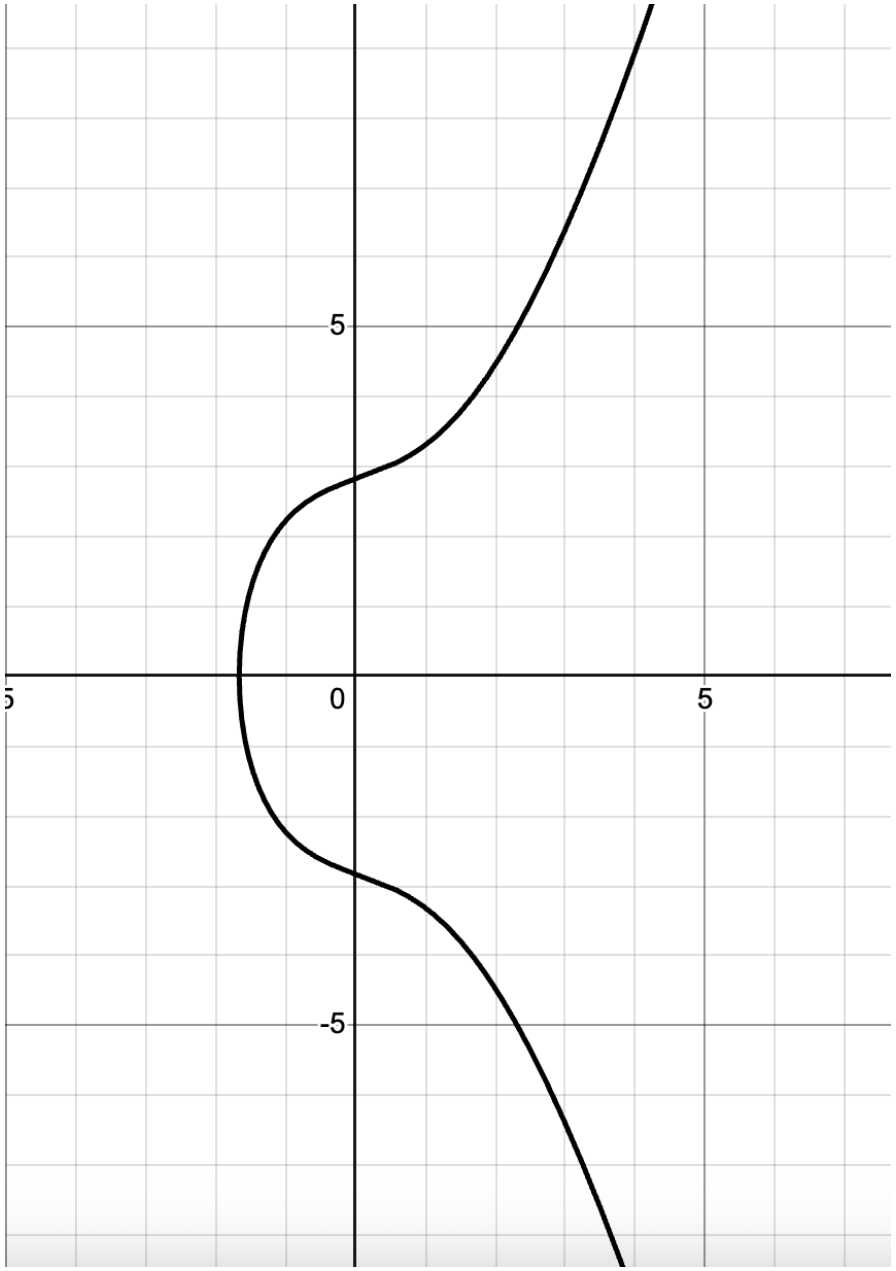
RareSkills

But if...



RareSkills

And if...



RareSkills

Elliptic curves forms cyclic groups...

$$\mathbb{E}_{a,b} = \{(x, y) \in \mathbb{F} \times \mathbb{F} | y^2 = x^3 + ax + b\} \cup \{\mathcal{O}\}$$

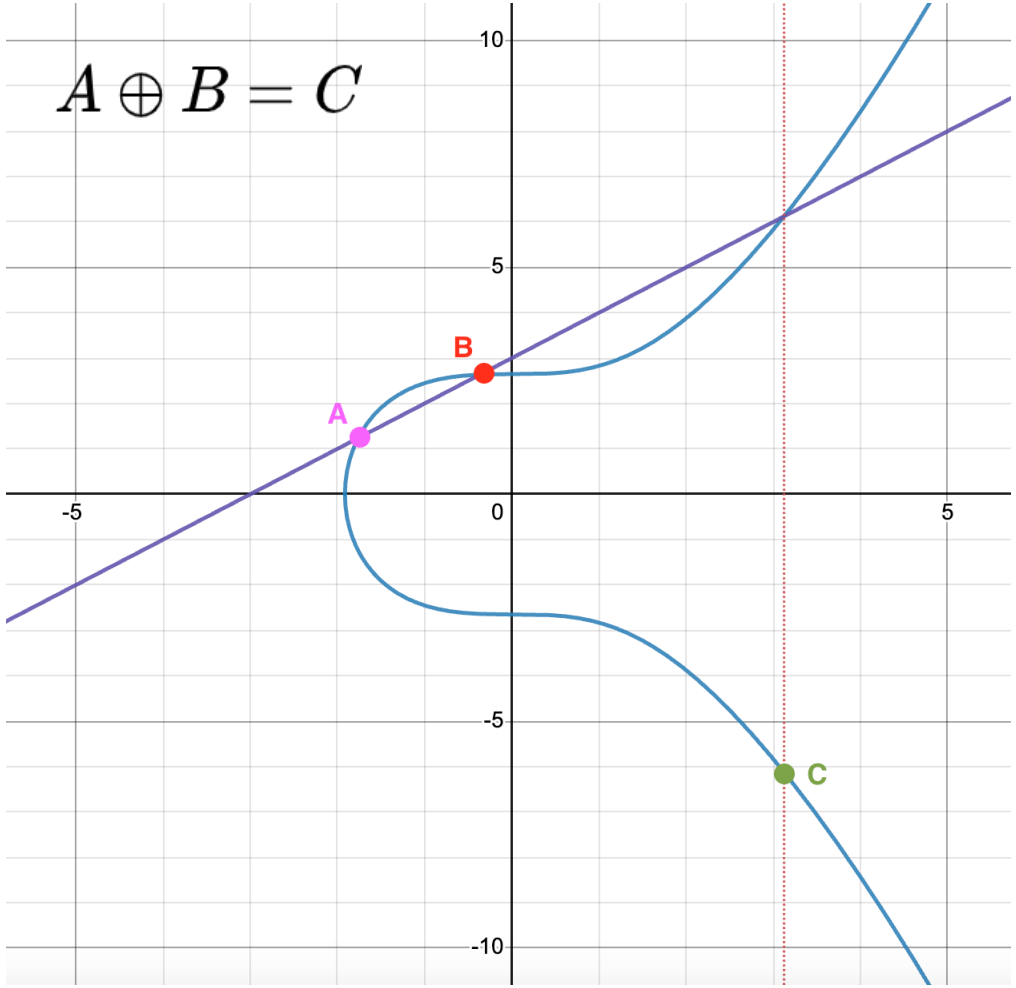
F is a field, O is the point at infinity

The operation is the **chord-and-tangent rule**

RareSkills

$$P_1 \neq P_2 \text{ and } x_1 \neq x_2$$

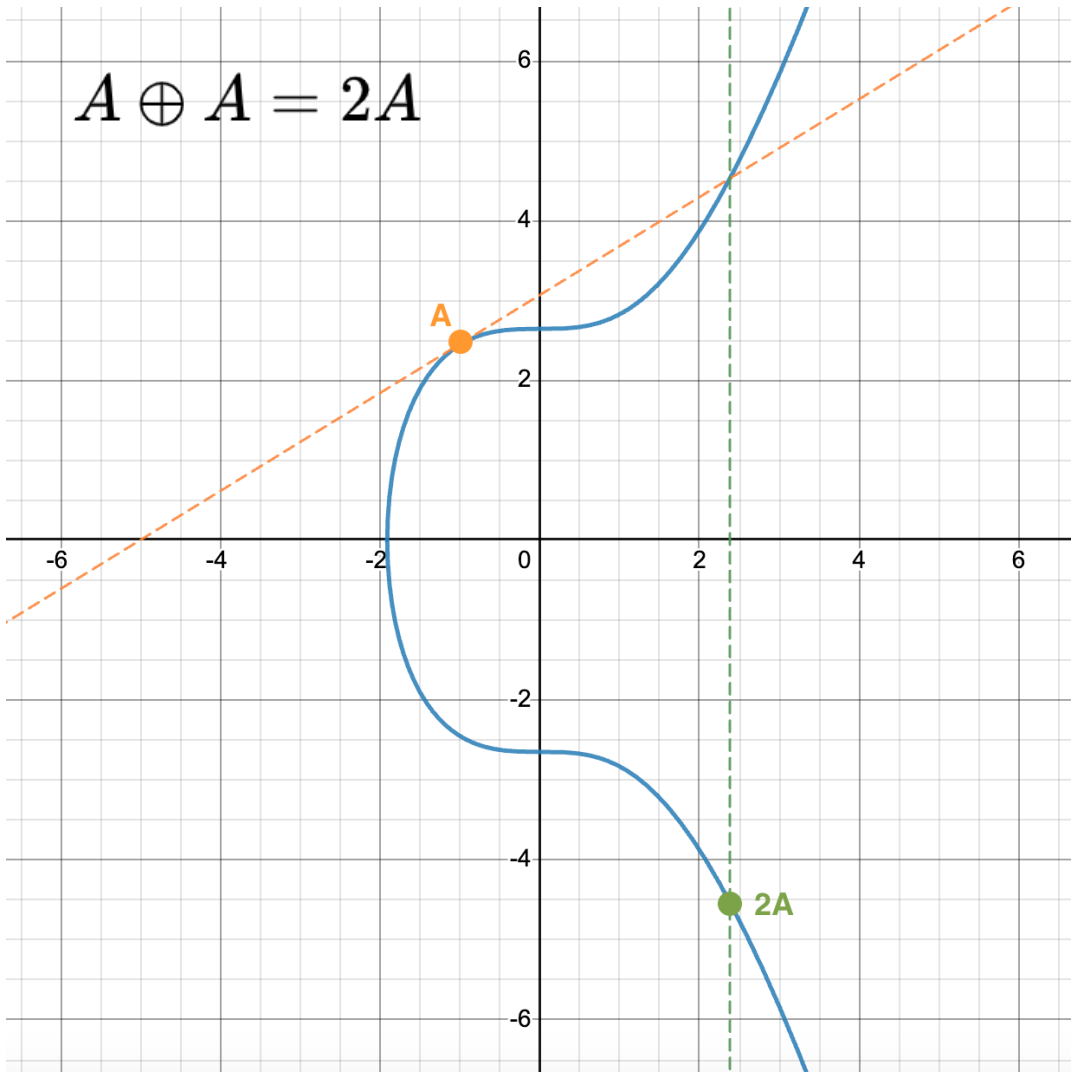
$$x_3 = \lambda^2 - x_1 - x_2$$
$$y_3 = \lambda(x_1 - x_3) - y_1$$
$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}$$



RareSkills

$$P_1 = P_2 \text{ and } x_1 \neq x_2$$

$$x_3 = \lambda^2 - 2x$$
$$y_3 = \lambda(x - x_3) - y$$
$$\lambda = \frac{3x^2 + a}{2y}$$



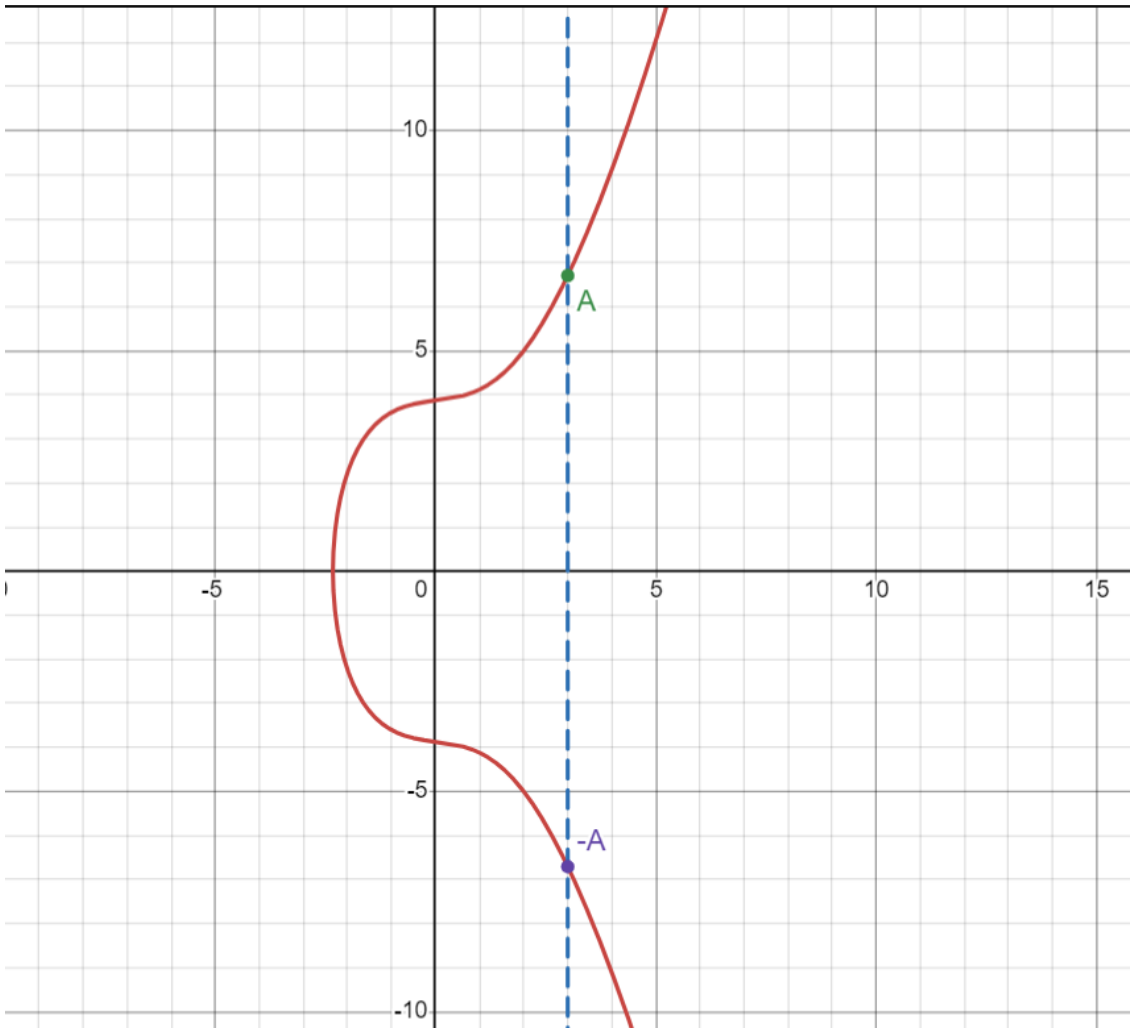
RareSkills

$$P_1 \neq P_2 \text{ and } x_1 = x_2$$

$$A \oplus -A = \mathcal{O}$$

$$\forall P \in \mathbb{E}_{a,b}$$

$$P \oplus \mathcal{O} = \mathcal{O} \oplus P = P$$



RareSkills

Why points flip?

RareSkills

An exponential map for EC

We can define a map $\mathbb{Z}_n : x \to [x]G$, where G is a generator of the group and $[x]G$ is $G \oplus G \oplus \ldots \oplus G$ x times.

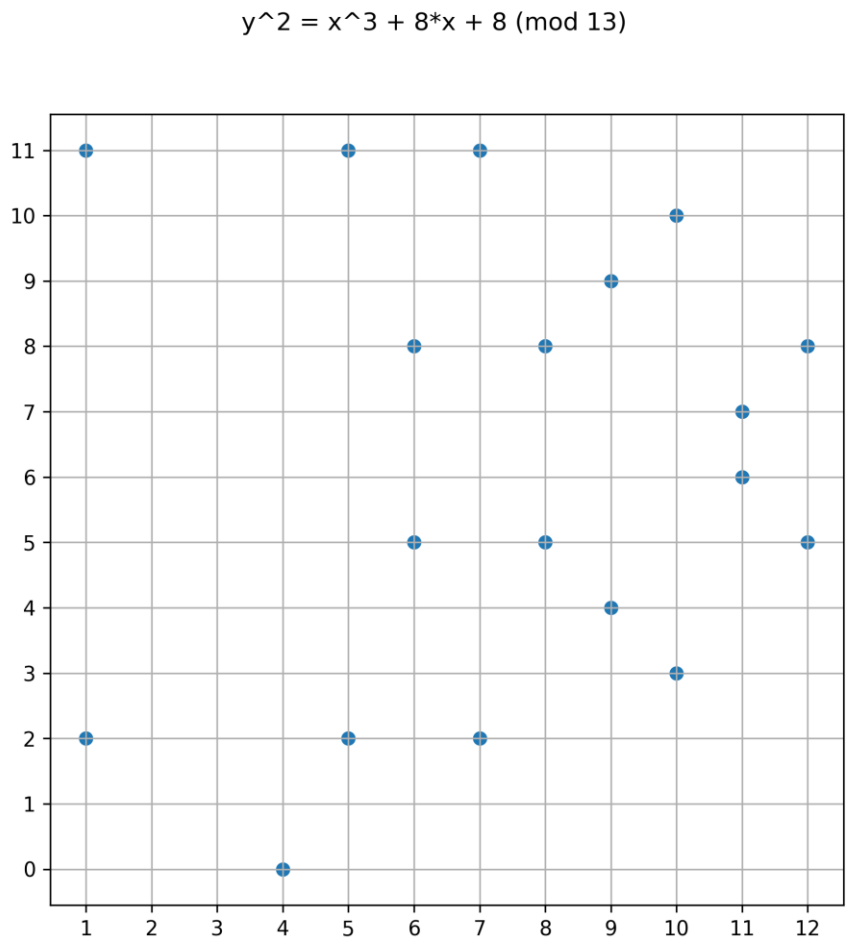
$$[x+y]G = [x]G \oplus [y]G$$

Example:

$$5G = 3G + 2G$$

RareSkills

Elliptic Curves in a finite field

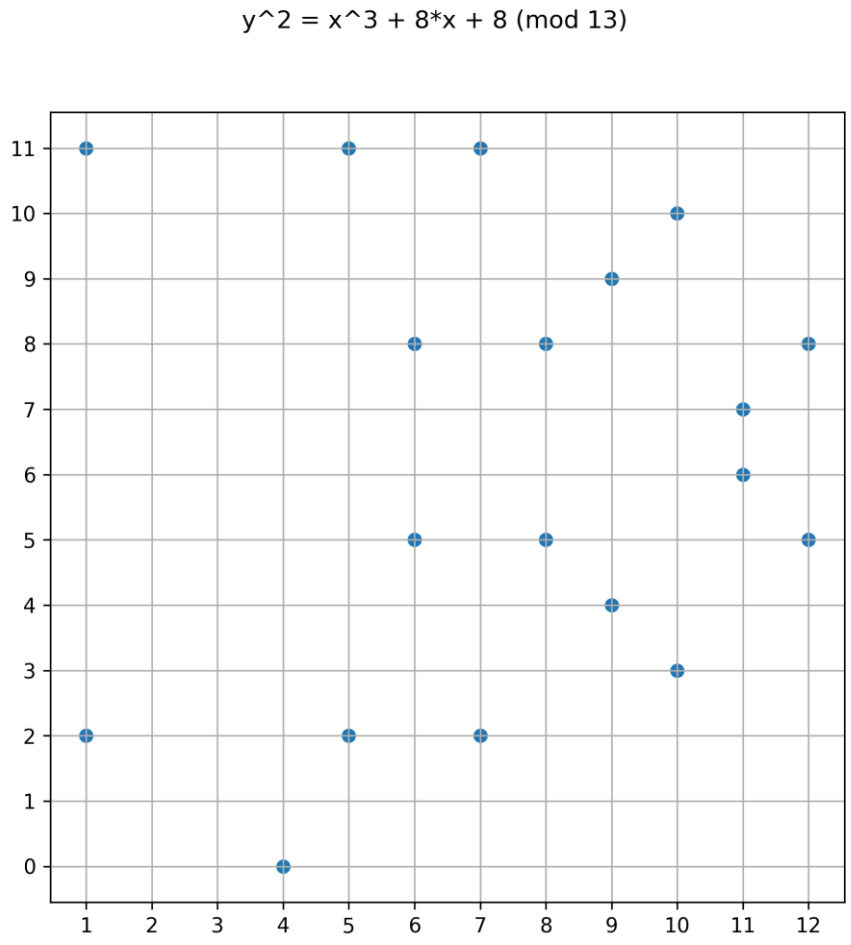


Import libnum

libnum.has_sqrtmod_prime_power(value, mod, 1)

libnum.sqrtmod_prime_power(value, mod,1)

Elliptic Curves in a finite field



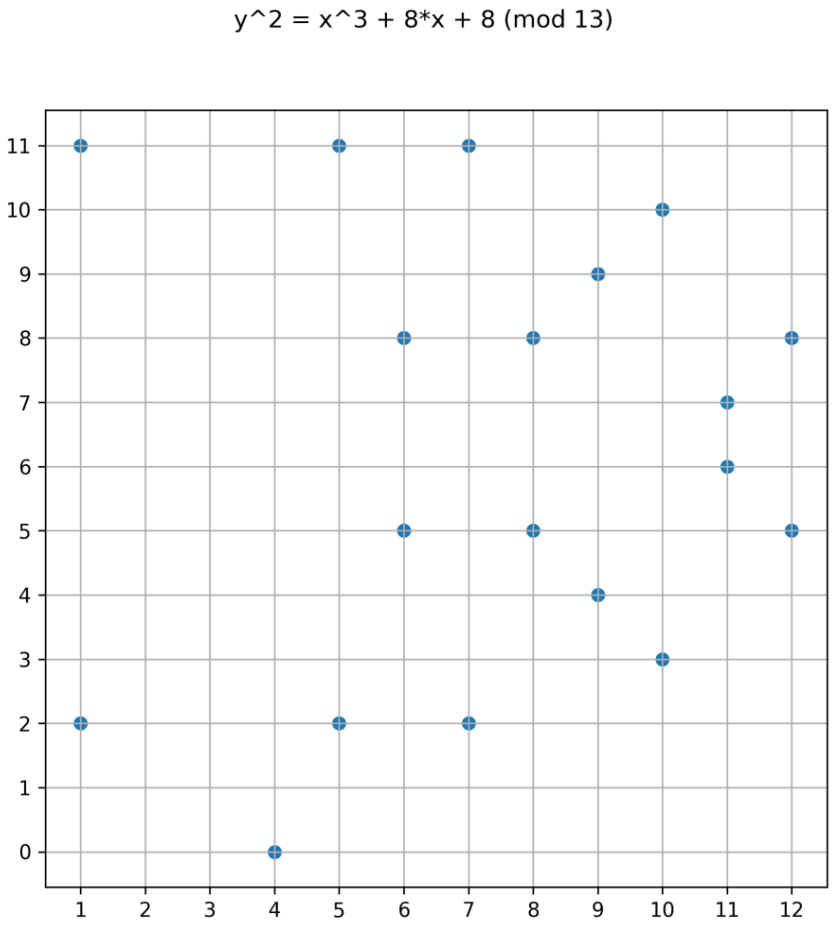
Order: number of elements of the cyclic group

$$\mathrm{Char}(\mathbb{F}_{13}) = 13$$

$$\mathrm{Order}(\mathbb{E}_{8,8}(\mathbb{F}_{13})) = 20$$

RareSkills

Example: adding points



RareSkills

The expontential map (or the homomorphism)

$$y^2 = x^3 + 8x + 8$$

$$\mathrm{Char}(\mathbb{F}_{13}) = 13$$

$$\mathrm{Order}(\mathbb{E}_{8,8}(\mathbb{F}_{13})) = 20$$

$$\mathbb{Z}_{20} \to \mathbb{E}_{8,8}(\mathbb{F}_{13})$$

RareSkills

ECDSA (homework)

- Setup: pick an elliptic curve and generator (secp256k1)
- Signer creates a public and private key: p, P = pG
- Signer hashes a message: h = hash("rareskills")
- Signer picks a random k and computes r = (kG).x
- Signer computes $s = k^{-1}(h + rp)$
- Signature is (P, h, r, s)
- Verifier computes $R' = s^{-1}(hG + rP)$
- If R'.x = r verifier accepts
- Why it works:
- $s^{-1}(hG + rP) = k(h + rp)^{-1}(hG + rP) = k(h + rp)^{-1}(hG + rpG) = \ldots$
- $k(h + rp)^{-1}(h + rp)G = k\cancel{(h + rp)^{-1}(h + rp)}G = kG$